

Bridging the Treatment Gap: A Novel LLM-Driven System for Scalable Initial Patient Assessments in Mental Healthcare

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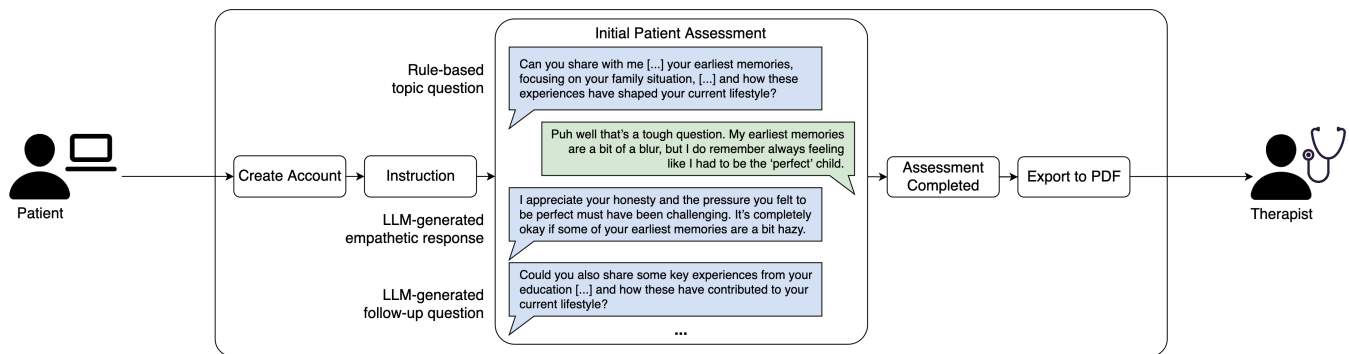


Figure 1: Proposed hybrid system for conversational agent-based initial patient assessments

Abstract

In mental healthcare, initial patient assessments function as the essential entry point to treatment. Due to the evident treatment gap in healthcare, solutions for optimizing efficiency and scalability of psychotherapy have to be found. This paper investigates an automated approach to initial patient assessments. Our proposed concept combines rule-based dialog management and topic coverage with the capabilities of large language models for dynamic conversations in a novel chatbot-based system. We hypothesize that automating patient evaluations can shorten the initial therapy phase, thereby increasing access and cost-effectiveness. To investigate the usability, user experience, and perceived feasibility of the system, we conducted a randomized controlled trial with 72 participants using an online form-based control. Additional qualitative feedback was collected through an expert interview. While

group differences in quantitative results are non-significant, qualitative feedback provides valuable insights into the potential and shortcomings of the approach and serves as a basis for future work.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Natural language interfaces*; • **Applied computing** → **Health informatics**; • **General and reference** → Empirical studies.

Keywords

Natural Language Processing, Internet-based Psychotherapy, Patient Assessment, Conversational Agents, Large Language Models, Usability

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1 Introduction

The Initial Patient Assessment (IPA) is the first step of psychotherapy, comprised of gathering essential information about the patient, building the foundation for the patient-therapist relationship, and assessing the patient's fit with the therapist and the form of treatment [12, 28]. A typical assessment consists of discussing the patient's biography, sexuality, self-perception, psychiatric medical history, addiction history, psychosocial information about the family, somatic history, forensic history, and conducting a third-party anamnesis with friends and family [13]. As the prevalence of mental health disorders increases [35] and treatment coverage is poor with only one mental health professional per 100,000 inhabitants in low-income countries and 60 per 100,000 in high-income countries [22], advancements making psychotherapy more scalable and cost-effective are required. Digital technologies may be able to support the stressed healthcare system [22], as research shows beneficial evidence for digital interventions [2, 23]. With rapid advancements in natural language generation through Large Language Models (LLMs) [11], great potential can be seen in streamlining healthcare practice [26], but the use of LLM services in healthcare also raises concerns related to false information, reproduction of biases, and data privacy [20]. While LLMs and their many applications have seen great focus from the research community, their use for IPAs in the context of psychotherapy has not been investigated yet. To assess the feasibility of using LLM-generated dialog for conducting IPAs, we introduce a novel chatbot-based system capable of autonomously performing assessments. Striving to minimize shortcomings associated with LLM use [20], we propose a hybrid design of both Artificial Intelligence (AI)-generated and rule-based dialog. Using this hybrid approach, the system can gather the most important information from the patient in a dynamic, LLM-driven conversation and can summarize collected information for the therapist while at the same time ensuring sufficient topic coverage through a rule-based algorithm. This can save valuable time for both patient and therapist allowing the patient to begin treatment faster. To evaluate the proposed system's usability, User Experience (UX), and feasibility, we conducted a randomized controlled trial with 72 participants. As our focus was on the general feasibility and acceptance of this novel concept, we conducted this first user study with healthy individuals and not with patients. While the experimental group used the novel hybrid approach, the control group used a classical form-based online survey on their medical history. Technological affinity, usability, UX, and adherence were measured using standardized questionnaires. Additionally, qualitative feedback was obtained through open-ended questions. An expert in internet-based psychological interventions was interviewed in a semi-structured interview to gather additional qualitative feedback.

This paper is structured as follows: Firstly, related work is investigated in Section 2. Secondly, the target audience, user requirements, a system concept, and implementation are presented in Section 3. Finally, the design and results of a conducted user study and expert interview are laid out (Section 4), discussed (Section 5), and a conclusion is drawn in Section 6.

2 Related Work

Research interest in the application of AI and LLMs in mental healthcare [10, 15] and in the use of chatbots for mental health support [19, 32] is growing rapidly. Especially in the context of online social networks, the application of Natural Language Processing (NLP) approaches for mental health assessments has been explored [1]. While the use of said technology for an IPA in mental healthcare has not been investigated yet, some research in related areas exists.

Williams et al. [36] investigated the use of LLMs for the assessment of clinical acuity of patients visiting the Emergency Department (ED) in a cross-sectional study of 251,401 adult ED visits. The authors extracted a balanced dataset of 10,000 patient pairs and used the gpt-4-0314 model by OpenAI in a zero-shot approach without in-context learning or fine-tuning. With an accuracy of 89% the LLM's performance was comparable with a human physician control indicating that the technology could be used to classify the acuity level between patients when provided with the medical history. Bantilan et al. [4] trained different AI and Machine Learning (ML) approaches on a dataset of 85,216 labeled patient sentences to assess the risk of suicide. Their best model was able to distinguish between suicide risk and non-risk content with an accuracy of 82.78%. Lepow et al. [18] used a lexical NLP approach to assess levels of suicidality severity by distinguishing between patients with suicidal ideation, suicide-related behavior, and suicide attempts based on notes taken during therapy sessions. Their approach showed an accuracy above 85% and performed better than the classification via International Statistical Classification of Diseases and Related Health Problems (ICD). Tanana et al. [33] compared different NLP models for recognizing sentiment in psychotherapy sessions. In their study, an LLM-based model using Bidirectional Encoder Representations from Transformers (BERT) outperformed traditional ML-based approaches. Striegl et al. [31] fine-tuned a BERT-based model for dimensional emotion recognition in the context of internet-based psychotherapy using Conversational Agents (CAs). Their approach reached an average Pearson correlation coefficient of 0.90 for valence and 0.77 for arousal on different test datasets. Lai et al. [16] fine-tuned an LLM on a collection of psychology Q&As as an answer generation and screening tool for therapists and patients. The authors evaluated the perplexity of the model, as well as the similarity and diversity of generated responses. Additionally, they conducted a pilot user study (N=6). Results showed the potential of the technology but did not reach human-level performance, as generated answers lacked logical coherence.

As the automatization of IPAs through LLMs and delivery via CAs has yet to be investigated, the following Section presents the design of a system for said purpose.

3 System Design

Following the human-centered design approach, we will first look at the target audience of a CA-based IPA system to derive user requirements and constraints which guide the system design. Afterward, we will outline the conceptual framework and implementation before moving on to its evaluation.

3.1 Conceptual Framework

The system’s target audience includes patients using the CA for mental health assessments and therapists working with the resulting data. Addressing their needs, our conceptual framework is guided by functional and non-functional requirements derived from existing studies [7, 24] and from an analysis of the context of use. Functional requirements include conducting an IPA through a hybrid CA, exporting results, and providing support for crisis situations [24]. Non-functional requirements emphasize accessibility, autonomy, and anonymity, alongside compliance with data privacy regulations (e.g., General Data Protection Regulation (GDPR)) and fostering empathetic interactions to build understanding and connection [7]. Building on these requirements, the conceptual framework integrates rule-based and LLM-driven components to balance structure, reliability, and consistency with flexibility and dynamic, empathetic interactions.

Overview of Conversation Flow. The conversation flow sequentially addresses the essential topics in IPAs (cf. Section 1). A welcome message and brief explanation of the assessment process is followed by topic-specific discussions. Each topic represents a critical area of the assessment and is addressed systematically to gather comprehensive and relevant information. Pre-defined questions introduce each topic, engaging users and maintaining focus. By evaluating user responses, the LLM facilitates follow-up questions dynamically to ensure sufficient coverage. Users can skip questions or terminate the interview at any point, respecting their autonomy and comfort. Figure 2 illustrates the conversation flow, highlighting the areas where the LLM assists.

Rule-Based vs. LLM-Driven Components. The hybrid approach combines the benefits of rule-based logic with LLM adaptability:

- **Rule-Based Components** enforce the conversation structure, ensure comprehensive topic coverage, manage user actions such as skipping questions or handling incomplete answers, and enable the system to react to crisis situations by detecting suicidal and self-harm ideation.
- **LLM-Driven Components** generate empathetic, context-aware responses to encourage user trust and engagement, analyze user inputs to determine completeness, and dynamically generate follow-up questions as needed. Furthermore, the LLM adapts the flow and order of topics based on the user’s prior responses for personalized and natural interactions, and summarizes patient inputs into concise, actionable outputs for therapists.

3.2 Implementation

Building on a pre-existing high-fidelity prototype [30] that provides essential functionality, the assessment system is implemented as a CA using the open-source, Python-based Rasa framework¹, which supports rule-based conversation structures. Rasa’s native NLP capabilities map user inputs to pre-defined intents, triggering actions such as initiating the assessment or skipping sensitive topics. An additional rule-based algorithm enforces the overall conversation

structure, ensuring all key assessment topics are addressed systematically. To ensure user safety, crisis situations are detected using a keyword-based approach and users are urged to call a linked hotline for mental health support in case self-harm or suicidal ideation is detected.

For more dynamic tasks, such as classifying user responses, generating empathetic replies, and formulating follow-up questions, Mistral AI’s Mixtral 8x7B [14] Mixture of Experts LLM is employed (as it had the best ratio between performance and required resources at the time of study in the category of open-source LLMs). Following a user message, these tasks are performed in a single API call, using a zero-shot prompt structured as context-task-format (see Listing 1). Iterative prompt engineering optimized performance and minimized token usage to reduce costs. A persistent conversation history is maintained using Rasa’s slot feature, allowing the LLM to access context and generate coherent responses across topics. Upon topic completion, the implemented rule-based algorithm marks topics as completed and stores relevant responses in slots. Dynamic topic sequencing is achieved by having the LLM select an uncompleted topic based on the prior conversation, creating a flexible yet structured interaction flow.

Listing 1: Prompt used for classifying user responses and crafting empathetic replies and follow-up questions

```
Context: conducting a medical history assessment
in the initial psychotherapy session. Does the
client response *roughly* satisfy the
question? Only superficially, don't expect
deep answers.

Question: '{assistant_question}'
Response: '{user_message}'

Answer only 'yes' or 'no'. Next, craft an
empathetic answer. Do not ask a question here.
Instead, only if not satisfied, also craft a
question afterwards leading user to provide
missing information.

Format:
`Satisfy: Yes/No
Answer: 'empathetic response'
Question: 'question'`
```

Privacy considerations are central to the system’s design. User inputs are anonymized using Microsoft Presidio², which detects personally identifiable information such as names, dates, and locations before further processing. Additionally, all data is encrypted after assessment completion and securely stored. A downloadable PDF, summarizing user responses and LLM-generated insights, is created client-side using jsPDF³, preserving data privacy by minimizing server-side processing.

To complement the CA and ensure parity in content, a control group system is implemented as a traditional form-based online survey using SurveyJS⁴. The form incorporates questions derived from an openly accessible anamnesis questionnaire [8], supplemented

¹Rasa - Open-source conversational AI, <https://opensource.rasa.com/>, accessed on 1st Jan '25

²microsoft/presidio: Data protection and de-identification SDK, <https://github.com/microsoft/presidio>, accessed on 6th Dec '24

³parallax/jsPDF: Client-side JavaScript PDF generation, <https://github.com/parallax/jsPDF>, accessed on 6th Dec '24

⁴SurveyJS, <https://surveyjs.io>, accessed on 6th Dec '24

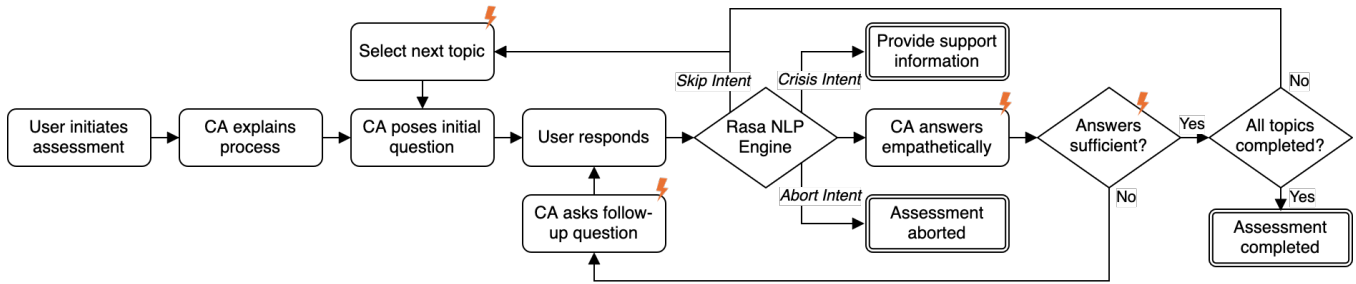


Figure 2: Concept of conversational flow, flashes mark LLM interactions

with additional topics to match the CA’s assessment scope. This ensures both systems address the same critical topics, allowing for a fair comparison of user experience and outcomes. Similar to the CA, form responses are encrypted, securely stored, and made exportable as PDFs.

4 User Study with Healthy Individuals

To assess the usability and UX of conducting an IPA using a hybrid CA combining rule-based and LLM-driven dialog, we conducted a randomized controlled trial, capturing both quantitative and qualitative feedback.

4.1 Methods and Materials

The user study was conducted remotely and unsupervised, comprising three stages. First, participants completed a pre-questionnaire via an anonymous online survey tool. This stage acquired demographic information and assessed participants' affinity with technology using the standardized Affinity for Technology Interaction (ATI) questionnaire [9]. Participants who reported being currently in therapy or having a prior mental health diagnosis were excluded to comply with ethical guidelines. Therefore, the study was conducted with healthy individuals and not with patients. All participants provided informed consent and signed a privacy policy in line with data protection regulations. Study results were fully anonymized. Sensitive data, provided to the system during assessments, was fully encrypted and deleted after the study. Data protection and privacy was monitored by the data protection officer of the University of Technology Dresden.

In the second stage, participants were redirected to the hosted system and randomized into two groups. The experimental group used the novel LLM-assisted CA, while the control group completed the traditional form-based online survey. Due to the extensive nature of IPAs, three topics - sexual history, addiction history, and forensic history - were excluded for both groups to shorten the time needed for participants to partake in the study.

In the final stage, participants were redirected to the survey tool and completed the System Usability Scale (SUS) [6] and the User Experience Questionnaire (UEQ) [17, 27] to evaluate usability and UX. Participants also answered four optional open-ended questions about their overall experience, helpful and frustrating features, and unmet needs. The study design was registered on OSF Registries [25].

Expert Feedback. To gain preliminary insights into the system’s clinical feasibility, an expert in internet-based psychological interventions participated in a semi-structured interview. Given an exported PDF with an exemplary assessment summary generated by the system, the expert was asked six questions on potential benefits and concerns, clinical impact, and accuracy. The PDF was generated from a conversation between the CA and a fictitious user, due to ethical and data protection reasons. This feedback served as a complementary perspective to the participant data, addressing considerations beyond usability and UX.

4.2 Results

4.2.1 Participants. A total of 442 individuals accessed the survey through social media, direct contact, and a university mailing list targeting students. Of these, 144 participants completed the survey, where 71 completions were exclusions due to ethical reasons. The high number of non-completions is typical for online surveys conducted on a large sample [37], especially in a multi-stage study design, due to time commitment or lack of interest.

From the 73 full completions, one had to be excluded due to a nonsensical entry, while 72 participants aged 18 to 55 (median 24) successfully completed the study, with roughly 60% identifying as female, 35% as male, and 5% as diverse. Most participants were students (81.9%) predominantly from Germany (83.3%), with some from India (5.6%). In total, 39 participants completed the form-based assessment (control group), while 33 participants completed the CA-based assessment (experimental group). Figure 3 breaks down the recruitment of participants visually.

4.2.2 Quantitative Results. Familiarity with technology, measured through the ATI, averaged 3.96 on a continuous 6-point scale. The control group reported slightly lower ATI scores (mean=3.868, SD=0.800) compared to the experimental group (mean=4.111, SD=0.941), but differences were not statistically significant (Welch's T-test, $t = 1.219$, $p = 0.227$).

Participants using the novel CA-based system reported lower usability scores on the SUS, with a median score of 65, compared to the control group’s median score of 77.5 ($p = 0.005$), corresponding to a usability grade of D for the CA prototype according to Bangor, Kortum, and Miller [3]. Table 1 summarizes the SUS results for both groups.

UX, measured through the UEQ, showed mixed results, as visualized in Figure 4. While the experimental group scored higher in

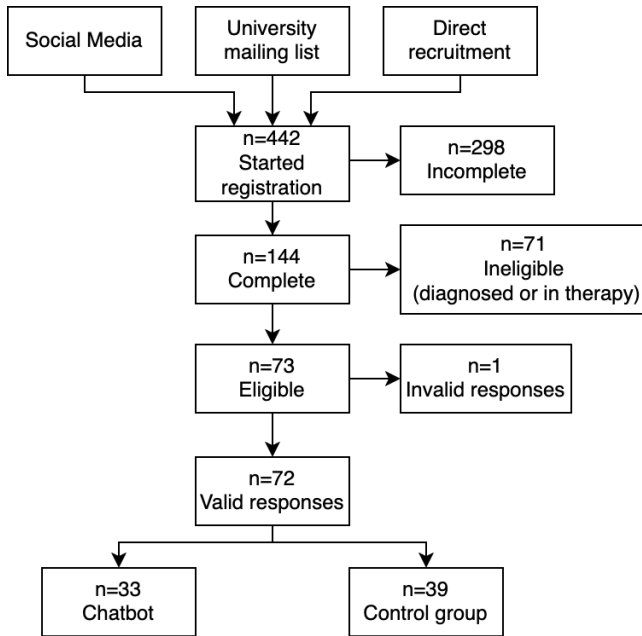
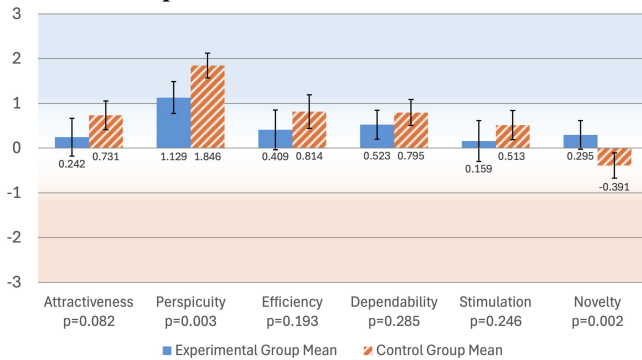


Figure 3: Breakdown of participant recruitment process

Table 1: Results of the SUS

SUS	Experimental Group	Control Group
N	33	39
Avg	64.697	75.897
Median	65	77.5
SD	15.522	16.498
Grade	D (OK)	C (Good)

Figure 4: Results of the UEQ visualized - Means, Confidence Intervals, and p-values



novelty (mean=0.295) compared to the control group (mean=-0.391, $p = 0.002$), they scored lower in perspicuity (mean=1.129 vs. 1.846, $p = 0.003$). Differences in other UX dimensions, including attractiveness, efficiency, dependability, and stimulation, slightly favored the control group but were not statistically significant.

4.2.3 Qualitative Results.

User Feedback. After the standardized questionnaires, participants optionally answered four open questions on their overall experience, helpful features, frustrating aspects, and unmet needs. Responses of the experimental group ($N=29$) were evaluated using thematic analysis guided by the approach outlined by Braun and Clarke [5] whereas responses by the control group ($N=34$) were not analyzed as the focus of this study lay in collecting formative feedback of the prototypical system for future system iterations. After systematically extracting codes, collating to themes, generalizing, and finally counting frequencies, responses revealed a mix of positive and negative perceptions.

Participants valued the system's reflective qualities ($N=9$), noting that it encouraged self-understanding and introspection. Some praised the chatbot's empathetic tone and personality ($N=4$) and its technical performance ($N=5$). A few described the experience as engaging and beneficial for organizing their thoughts. However, a subset of participants found no discernible advantages to the system ($N=4$).

Negative impressions centered around usability and interaction challenges. Repetitive questions ($N=22$) were a predominant source of frustration, leading to perceptions of redundancy and inefficiency. Additionally, participants found the question phrasing overly complex ($N=9$), contributing to feelings of annoyance ($N=6$) and time consumption ($N=3$). Some expressed discomfort sharing personal details with an AI system ($N=3$).

Participants highlighted several unmet needs. Suggestions included adding features such as advice or a diagnosis following the assessment ($N=4$), greater user control over topics and question depth ($N=3$), and more concise and direct questioning ($N=2$).

Expert Interview. The expert in internet-based psychological interventions identified several strengths in the system, emphasizing its potential to save time. The structured, rule-based approach augmented by LLMs was noted as a key benefit, ensuring consistency and reducing variability in assessments across therapists. Additionally, the system's capacity to generate empathetic responses and guide users toward self-reflection was highlighted as a positive step in enhancing patient engagement. The expert acknowledged that such innovations could alleviate cognitive load for therapists, enabling them to focus on deeper therapeutic connections during subsequent sessions.

Despite its strengths, the expert expressed concerns about the phrasing and complexity of the system's questions, suggesting that their formal tone might hinder natural interaction and rapport-building. Questions appeared overly generalized or abstract, contrasting with the conversational style typically employed in clinical practice. Moreover, legal and ethical challenges, such as accountability for errors in the AI's outputs, were raised as critical considerations. The expert recommended iterative testing, beginning with simpler questions and gradually integrating semi-structured components, to enhance the system's relevance and acceptability in clinical settings.

5 Discussion

The results of the user study indicate that users did not prefer the hybrid approach over traditional form-based assessment in UX and usability, though user and expert feedback showed the system's potential. While the use of CAs in healthcare is still in its infancy, we can derive various propositions.

Navigating Usability and UX in Initial Patient Assessments. The study revealed neutral UX for both the novel system and the control group (cf. Figure 4), establishing a solid foundation for further improvements. Pre-assessment questionnaires are often perceived as tedious [21, 34], potentially creating barriers to therapy access [29]. This challenge is particularly prominent for users without an immediate need for therapy, likely contributing to the neutral UX observed across both groups. While the system demonstrated potential for fostering reflection and empathetic interactions, it fell short in meeting user expectations for efficiency and clarity. These shortcomings were reflected in the SUS scores, which, in combination with the UEQ, effectively captured quantitative measures but lacked nuance. Open-ended questions revealed additional positive aspects and contextualized the scores, revealing the system's promise to support introspection and engage users meaningfully. These insights suggest that, with refinement, the system could address gaps in existing assessment methods and enhance the overall experience for both patients and therapists.

Refining the Approach. The hybrid system's repetitive conversations and occasional context loss caused frustration (cf. Section 4.2.3). However, these challenges were not due to inherent flaws in the hybrid approach but rather stemmed from the difficulty of balancing rule-based precision with LLM flexibility in this first system iteration. Limitations in the prototype, including minimal iteration cycles during development and the use of a weaker LLM with limited context retention, likely contributed to these issues.

To address these challenges, future systems combining rule-based with LLM-generated dialog should prioritize UX and explore novel LLM approaches, such as Retrieval Augmented Generation (RAG) and LLM-based autonomous agents [38]. In future iterations, RAG could be used to dynamically retrieve clinically validated resources, ensuring contextually accurate and relevant follow-ups during the conversation. Similarly, autonomous agents and the utilization of newer LLMs could enhance the system's adaptability by retaining key context across topics and offering a more natural interaction flow. Nevertheless, the hybrid approach created the possibility for a dynamic conversational flow with follow-up questions and empathic answers while at the same time ensuring topic coverage through a rule-based algorithm and user safety through keyword-based detection of suicidal and self-harm ideation. The benefits of the approach were reflected in qualitative feedback, as the system's reflective and empathetic qualities, the encouragement for introspection, and the insurance of consistency were positively mentioned by both participants and the interviewed expert.

Interdisciplinary Collaboration is Essential. This study highlights the importance of interdisciplinary collaboration in system design and content creation. Both the expert and users expressed dissatisfaction with system reliability, question complexity, and wording.

Psychologists contribute expertise in phrasing empathetic, clinically valid questions, while UX designers can assist in crafting engaging and accessible interfaces and technical experts ensure system reliability. By fostering collaboration, future iterations can align more closely with ethical and clinical standards, particularly in sensitive domains like mental health.

Participant Dropout and Group Disparity. The algorithmically randomized group assignment resulted in a higher completion rate for the control group compared to the experimental group. This discrepancy likely stems from the time-consuming and repetitive nature of interacting with the CA in the experimental group, as reported in the user feedback (cf. Section 4.2.3). In contrast, the control group used a familiar online survey format with a predictable structure of questions and a progress indicator, which helped users know what to expect. In the experimental group, participants expressed frustration over the lack of transparency about the interview's length, with some wishing for an overview of the questions being asked and complaining about the length exceeding the advertised time. However, no correlation was found between technical affinity and dropouts, suggesting that the challenges were due to the system design rather than the demographics. To address these issues, future versions should provide users with clear expectations about the process, including progress indicators and estimated completion times, to reduce frustration and increase engagement.

Ethical Considerations of the Presented Approach. While novel solutions are needed to fasten processes in mental healthcare, the Initial Patient Assessment is a critical part of therapy. Thus, the application of AI in this field should always be treated with caution. Developed systems should always be used with a human-in-the-loop approach employing human mental health professionals to diagnose and triage patients. Thus, generated reports of CA-based initial patient assessments should only be utilized to fasten the initial patient evaluation by a human clinical expert. Due to the personal nature of information shared during the assessment, data privacy and protection have to be considered. Furthermore, certain questions during the assessment might evoke negative feelings and mental health crises. This stresses the need for safety mechanisms and further motivates a hybrid approach that includes a reliable, rule-based component while also leveraging the capabilities of Large Language Models.

Limitations. This study represents an early exploration of a novel hybrid approach to CAs for IPA, marking a first iteration of the human-centered design cycle. As a first-pass prototype, the system is not yet fully developed, with limitations such as untested conversation scenarios, occasional repetitiveness, and constrained context retention due to reliance on a weaker LLM. These shortcomings highlight significant opportunities for refinement, where expert involvement during development could further enhance user understanding and ground the system in practical clinical experience. The small sample size and narrow demographic composition limit the generalizability of findings. Furthermore, the study was conducted with healthy participants and one expert interview only, to gain preliminary insights into its feasibility, acceptance, and UX before further developing the system in a next iteration. After refining the system based on the collected formative feedback, future studies

should recruit participants with immediate therapy needs to evaluate the system's relevance and usability in real-world scenarios. Additionally, expanding the sample size, demographic diversity, and the number of experts in subsequent studies will ensure broader applicability. These early findings establish a foundation for future iterations, where advancements in UX and closer alignment to clinical practices could be achieved.

6 Conclusion

This study explored the potential of a hybrid AI system combining rule-based and LLM-driven dialog for CA-based IPA in mental healthcare. While the proposed system did not achieve significant improvements in UX or usability compared to traditional form-based assessment, qualitative feedback highlighted its unique strengths. Users appreciated its ability to facilitate reflection, introspection, and empathetic interactions, suggesting value beyond the measured quantitative outcomes. Our findings further underpin the inherent challenges of making such interaction parts of therapies engaging and efficient. This highlights the need for innovative approaches in Human-Computer Interaction (HCI) that combine the rigor and methodologies of serious applications with the engaging and playful aspects of interaction design to create more compelling user experiences.

Our findings emphasize the need for interdisciplinary collaboration in designing and developing mental health systems that meet clinical needs, align with user expectations, and adhere to ethical standards. While the use of AI systems in mental healthcare holds great potential, their implementation must involve human experts in critical decisions such as triaging, diagnosing, or determining a patient's fit for therapy. AI-based tools, including the one presented in this paper, should be viewed as supportive technologies that complement, rather than replace, human-driven interventions and assessments.

Future work should prioritize testing with the actual target audience – individuals in immediate need of therapy – to better evaluate the system's practical impact. The integration of emerging technologies, such as RAG and autonomous AI agents, offers promising directions for improving retention, reduced repetition, and enhanced user engagement. By addressing current limitations, refining UX, and enhancing clinical validity, this approach holds the potential to make IPA more accessible, increase access to treatment through more cost-effective assessments, and reduce waiting times.

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